MATH 181A - Homework 4

due Saturday, January 31 at 11:59PM

Problem 1. 5.4.19

Let $Y_1, Y_2, \ldots, Y_n$ be a random sample of size n from the pdf $f_Y(y;\theta) = \frac{1}{\theta}e^{-y/\theta}, y > 0$.

1.1 Show that $\hat{\theta}_1 = Y_1, \hat{\theta}_2 = \bar{Y}$, and $\hat{\theta}_3 = n \cdot Y_{\min}$ are all unbiased estimators for θ.

Solution: Since $Y_i \sim$ Exponential(θ), we have $\mathbb{E}[\hat{\theta}_1] = \mathbb{E}[Y_1] = \mathbb{E}[Y_i] = \theta$. We also have $\mathbb{E}[\hat{\theta}_2] = \mathbb{E}[\bar{Y}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n Y_i\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[Y_i] = \frac{1}{n}\sum_{i=1}^n \theta = \theta$. Hence, $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators for θ.

For $\hat{\theta}_3 = n \cdot Y_{\min}$, we have $\mathbb{E}[\hat{\theta}_3] = \mathbb{E}[n \cdot Y_{\min}] = n \cdot \mathbb{E}[Y_{\min}]$. We know that $Y_{\min}$ is the minimum of n i.i.d. Exponential random variables with parameter θ, so its distribution is:

$$P(Y_{\min} \geq y) = \prod_{i=1}^n P(Y_i \geq y) = \prod_{i=1}^n e^{-y/\theta} = e^{-ny/\theta}, \quad F_{Y_{\min}}(y) = 1 - e^{-ny/\theta}$$

$$f_{Y_{\min}}(y) = \frac{d}{dy}F_{Y_{\min}}(y) = \frac{d}{dy}(1 - e^{-ny/\theta}) = \frac{n}{\theta}e^{-ny/\theta}$$

Now we compute $\mathbb{E}[Y_{\min}]$ using IBP with $u = y$ and $dv = \frac{n}{\theta}e^{-ny/\theta}dy$ to find:

$$\mathbb{E}[Y_{\min}] = \int_0^\infty y\frac{n}{\theta}e^{-ny/\theta}dy = \left[-ye^{-ny/\theta}\right]_0^\infty + \int_0^\infty e^{-ny/\theta}dy = \left[-\frac{\theta}{n}e^{-ny/\theta}\right]_0^\infty = \frac{\theta}{n}$$

Thus, we have $\mathbb{E}[\hat{\theta}_3] = n \cdot \mathbb{E}[Y_{\min}] = n \cdot \frac{\theta}{n} = \theta$. Hence, $\hat{\theta}_3$ is an unbiased estimator for θ.

1.2 Find the variances of $\hat{\theta}_1, \hat{\theta}_2,$ and $\hat{\theta}_3$.

Solution: Since $Y_i \sim$ Exponential(θ) we know that $\text{Var}(Y_i) = \theta^2$. Thus, we have $\text{Var}(\hat{\theta}_1) = \text{Var}(Y_1) = \text{Var}(Y_i) = \theta^2$. Now, for $\hat{\theta}_2 = \bar{Y}$, we have:

$$\text{Var}(\bar{Y}) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^n Y_i\right) = \frac{1}{n^2}\text{Var}\left(\sum_{i=1}^n Y_i\right) = \frac{1}{n^2}\sum_{i=1}^n \text{Var}(Y_i) = \frac{1}{n^2}\sum_{i=1}^n \theta^2 = \frac{\theta^2}{n}$$

Finally, for $\hat{\theta}_3 = n \cdot Y_{\min}$, we have:

$$\mathbb{E}[Y_{\min}^2] = \int_0^\infty y^2 \cdot \frac{n}{\theta}e^{-ny/\theta}dy = \left[-y^2e^{-ny/\theta}\right]_0^\infty + \int_0^\infty 2ye^{-ny/\theta}dy = 0 + 2\frac{\theta}{n}(\mathbb{E}[Y_{\min}]) = 2\frac{\theta^2}{n^2}$$

Thus, we have $\text{Var}(\hat{\theta}_3) = n^2\text{Var}(Y_{\min}) = n^2(\mathbb{E}[Y_{\min}^2] - \mathbb{E}[Y_{\min}]^2) = n^2(2\frac{\theta^2}{n^2} - \frac{\theta^2}{n^2}) = \theta^2$.

1.3 Calculate the relative efficiencies of $\hat{\theta}_1$ to $\hat{\theta}_3$ and $\hat{\theta}_2$ to $\hat{\theta}_3$.

Solution: Using $\text{Var}(\hat{\theta}_1) = \theta^2$, $\text{Var}(\hat{\theta}_2) = \frac{\theta^2}{n}$, and $\text{Var}(\hat{\theta}_3) = \theta^2$, we have: $RE(\hat{\theta}_1, \hat{\theta}_3) = \frac{\theta^2}{\theta^2} = 1$ and $RE(\hat{\theta}_2, \hat{\theta}_3) = \frac{\theta^2}{\frac{\theta^2}{n}} = n$. Hence, $\hat{\theta}_1$ and $\hat{\theta}_3$ are equally efficient, and $\hat{\theta}_2$ is n times more efficient than $\hat{\theta}_1$ and $\hat{\theta}_3$.

Problem 2. 5.4.20

Given a random sample of size n from a Poisson distribution, $\hat{\lambda}_1 = X_1$ and $\hat{\lambda}_2 = \bar{X}$ are two unbiased estimators for λ. First show that the two estimators in the problem are unbiased, then calculate the relative efficiency of $\hat{\lambda}_1$ to $\hat{\lambda}_2$.

Solution: Since $X_i \sim \text{Poisson}(\lambda)$ we know that $\mathbb{E}[X_i] = \text{Var}(X_i) = \lambda$. Thus, we have $\mathbb{E}[\hat{\lambda}_1] = \mathbb{E}[X_1] = \mathbb{E}[X_i] = \lambda$. Next, we have:

$$\mathbb{E}[\hat{\lambda}_2] = \mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}\sum_{i=1}^{n}\lambda = \lambda$$

Hence, $\hat{\lambda}_1$ and $\hat{\lambda}_2$ are unbiased estimators for λ. Now, we know that $\text{Var}(\hat{\lambda}_1) = \text{Var}(X_1) = \lambda$, next:

$$\text{Var}(\hat{\lambda}_2) = \text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\text{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}(X_i) = \frac{1}{n^2}\sum_{i=1}^{n}\lambda = \frac{\lambda}{n}$$

Thus, we have $RE(\hat{\lambda}_1, \hat{\lambda}_2) = \frac{\frac{\lambda}{n}}{\lambda} = \frac{1}{n}$. Hence, $\hat{\lambda}_2$ is n times more efficient than $\hat{\lambda}_1$.

Problem 3. 5.4.22

Suppose that W_1 is a random variable with mean μ and variance σ_1^2 and W_2 is a random variable with mean μ and variance σ_2^2. From Example 5.4.3, we know that $cW_1 + (1-c)W_2$ is an unbiased estimator of μ for any constant $c > 0$. If W_1 and W_2 are independent, for what value of c is the estimator $cW_1 + (1-c)W_2$ most efficient?

Solution:

$$\text{Var}(cW_1 + (1-c)W_2) = c^2\text{Var}(W_1) + (1-c)^2\text{Var}(W_2) + 2c(1-c)Cov(W_1, W_2) = c^2\sigma_1^2 + (1-c)^2\sigma_2^2$$

Thus to minimize this value, we find where $\frac{d}{dc}\text{Var}(cW_1 + (1-c)W_2) = 0$.

$$\frac{d}{dc}\text{Var}(cW_1 + (1-c)W_2) = 2c\sigma_1^2 - 2(1-c)\sigma_2^2 = 0 \implies c = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

Thus, the value of c that minimizes the variance of $cW_1 + (1-c)W_2$ is $c = \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}$.

Problem 4.

Suppose $X_1, \ldots, X_n$ are i.i.d. random variables with $E[X_i] = \theta$ and $\text{Var}(X_i) = \sigma^2$. Let $\hat{\theta}_1 = \bar{X}$. Suppose $\hat{\theta}_2$ is another estimator based on $X_1, \ldots, X_n$, which is known to be unbiased for θ. Assume that $\text{Var}(\hat{\theta}_2) = \text{Var}(\hat{\theta}_1)/2$ and $Cov(\hat{\theta}_1, \hat{\theta}_2) = \text{Var}(\hat{\theta}_1)/3$. Consider the estimator $\hat{\theta} = c\hat{\theta}_1 + (1-c)\hat{\theta}_2$, where $c \in [0, 1]$.

4.1 Calculate the bias, variance, and mean squared error of $\hat{\theta}$. Hint: You shall use the fact $Cov(aX, bY) = abCov(X, Y)$ for any random variables X, Y and real numbers a, b.

Solution: Since $\hat{\theta}_1 = \bar{X}$ we know that $\mathbb{E}[\hat{\theta}_1] = \mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n}\sum_{i=1}^n \theta = \theta$. Thus, $\hat{\theta}_1$ is an unbiased estimator for θ. Now, we have:

$$\text{Var}(\hat{\theta}_1) = \text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^n X_i\right) = \frac{1}{n^2}\text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2}\sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2}\sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}$$

Thus, we have $\text{Var}(\hat{\theta}_1) = \frac{\sigma^2}{n}$. Now, for $\hat{\theta}_2$, we have $\text{Var}(\hat{\theta}_2) = \text{Var}(\hat{\theta}_1)/2 = \frac{\sigma^2}{2n}$. Now, we find the bias of $\hat{\theta}$:

$$\mathbb{E}[\hat{\theta}] = \mathbb{E}[c\hat{\theta}_1 + (1-c)\hat{\theta}_2] = c\mathbb{E}[\hat{\theta}_1] + (1-c)\mathbb{E}[\hat{\theta}_2] = c\theta + (1-c)\theta = \theta$$

Thus, we have $\mathbb{E}[\hat{\theta}] = \theta$ so the bias is 0 and $\hat{\theta}$ is unbiased. Now, we find the variance of $\hat{\theta}$:

$$\text{Var}(\hat{\theta}) = \text{Var}(c\hat{\theta}_1 + (1-c)\hat{\theta}_2) = c^2\text{Var}(\hat{\theta}_1) + (1-c)^2\text{Var}(\hat{\theta}_2) + 2c(1-c)Cov(\hat{\theta}_1, \hat{\theta}_2)$$

$$= c^2\frac{\sigma^2}{n} + (1-c)^2\frac{\sigma^2}{2n} + 2c(1-c)\frac{\sigma^2}{3n} = \frac{\sigma^2}{6n}(5c^2 - 2c + 3)$$

Now, we find the mean squared error of $\hat{\theta}$ as

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}^2 = \text{Var}(\hat{\theta}) + 0^2 = \text{Var}(\hat{\theta}) = \frac{\sigma^2}{6n}(5c^2 - 2c + 3)$$

4.2 For what value of c is the mean squared error of $\hat{\theta}$ the smallest?

Solution: To find the value of c that minimizes the mean squared error of $\hat{\theta}$, we find where $\frac{d}{dc}MSE(\hat{\theta}) = 0$.

$$\frac{d}{dc}MSE(\hat{\theta}) = \frac{d}{dc}\left(\frac{\sigma^2}{6n}(5c^2 - 2c + 3)\right) = 0 \implies 10c - 2 = 0 \implies c = \frac{1}{5}$$

Thus, the value of c that minimizes the mean squared error of $\hat{\theta}$ is $c = \frac{1}{5}$.

Problem 5. 5.5.1

Let $Y_1, Y_2, \ldots, Y_n$ be a random sample from the pdf $f_Y(y;\theta) = \frac{1}{\theta}e^{-y/\theta}, y > 0$. Compare the Cramer-Rao lower bound for $f_Y(y;\theta)$ to the variance of the maximum likelihood estimator for θ, $\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} Y_i$. Is $\bar{Y}$ a best estimator for θ?

Solution: Since $Y_i \sim \text{Exponential}(\theta)$ we know that $\mathbb{E}[Y_i] = \theta$ and $\text{Var}(Y_i) = \theta^2$. Thus, we have:

$$\text{Var}(\hat{\theta}) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2}\text{Var}\left(\sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}(Y_i) = \frac{1}{n^2}\sum_{i=1}^{n}\theta^2 = \frac{\theta^2}{n}$$

Now, we find the Fisher information:

$$I(\theta) = -\mathbb{E}\left[\frac{d^2}{d\theta^2}\log f_Y(y;\theta)\right] = -\mathbb{E}\left[\frac{d^2}{d\theta^2}\left(-\frac{y}{\theta} - \log\theta\right)\right] = -\mathbb{E}\left[-\frac{2y}{\theta^3} + \frac{1}{\theta^2}\right] = \frac{2}{\theta^2} - \frac{1}{\theta^2} = \frac{1}{\theta^2}$$

Thus, we have $I(\theta) = \frac{1}{\theta^2}$. Now, we find the Cramer-Rao lower bound:

$$CRLB = \frac{1}{nI(\theta)} = \frac{\theta^2}{n}$$

Thus, we have $CRLB = \frac{\theta^2}{n}$. Since $\text{Var}(\hat{\theta}) = \frac{\theta^2}{n} = CRLB$, we have that $\hat{\theta}$ is a best estimator for θ.

Problem 6. 5.5.2

Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from the Poisson distribution, $p_X(k;\lambda) = e^{-\lambda}\lambda^k/k!$, $k = 0, 1, \ldots$. Show that $\hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is an efficient estimator for λ.

Solution: Since $X_i \sim \text{Poisson}(\lambda)$ we know that $\mathbb{E}[X_i] = \text{Var}(X_i) = \lambda$. Thus, we have:

$$\text{Var}(\hat{\lambda}) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\text{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}(X_i) = \frac{1}{n^2}\sum_{i=1}^{n}\lambda = \frac{\lambda}{n}$$

Now, we find the Fisher information:

$$I(\lambda) = -\mathbb{E}\left[\frac{d^2}{d\lambda^2}\log p_X(k;\lambda)\right] = -\mathbb{E}\left[\frac{d^2}{d\lambda^2}\left(-\lambda + k\log\lambda - \log k!\right)\right] = -\mathbb{E}\left[-\frac{k}{\lambda^2}\right] = \frac{1}{\lambda}$$

Thus, we have $I(\lambda) = \frac{1}{\lambda}$. Now, we find the Cramer-Rao lower bound:

$$CRLB = \frac{1}{nI(\lambda)} = \frac{\lambda}{n}$$

Thus, we have $CRLB = \frac{\lambda}{n}$. Since $\text{Var}(\hat{\lambda}) = \frac{\lambda}{n} = CRLB$, we have that $\hat{\lambda}$ is an efficient estimator for λ.

Problem 7.

Let $X_1, \ldots, X_n$ be a random sample from the density function $f_X(x;\theta) = (\theta+1)x^\theta$ for $0 < x < 1$, and $f_X(x;\theta) = 0$ otherwise, where $\theta > -1$. Find the 100(1 - α)% confidence interval for θ based on the maximum likelihood estimator.

Solution: First we find the MLE of θ:

$$\ell(\theta) = \sum_{i=1}^{n} \log f_X(x_i;\theta) = \sum_{i=1}^{n} \log\left((\theta+1)x_i^\theta\right) = \sum_{i=1}^{n} \log(\theta+1) + \theta \log x_i$$

$$\frac{d}{d\theta}\ell(\hat{\theta}) = \frac{n}{\hat{\theta}+1} + \sum_{i=1}^{n} \log x_i = 0 \implies \hat{\theta} = -\frac{n}{\sum_{i=1}^{n} \log x_i} - 1$$

Now, we find the Fisher information:

$$I(\theta) = -\mathbb{E}\left[\frac{d^2}{d\theta^2} \log f_X(x;\theta)\right] = -\mathbb{E}\left[\frac{d^2}{d\theta^2} \left(\log(\theta+1) + \theta \log x\right)\right] = -\mathbb{E}\left[-\frac{1}{(\theta+1)^2}\right] = \frac{1}{(\theta+1)^2}$$

Now, the 100(1 - α)% confidence interval for $\hat{\theta}$ is given by:

$$\hat{\theta} \pm z_{\alpha/2}\sqrt{\frac{1}{nI(\hat{\theta})}} = \hat{\theta} \pm z_{\alpha/2}\frac{\hat{\theta}+1}{\sqrt{n}}$$